MULUNGUSHI UNIVERSITY

SCHOOL OF ENGEERING AND TECHNOLOGY

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COURSE NAME:DATA STRUCTURES AND ALGORITHMS

COURSE CODE: ICT 202

TOPIC: STACKS

ASSIGNMENT NUMBER:3

LECTURER: MR KALUBA

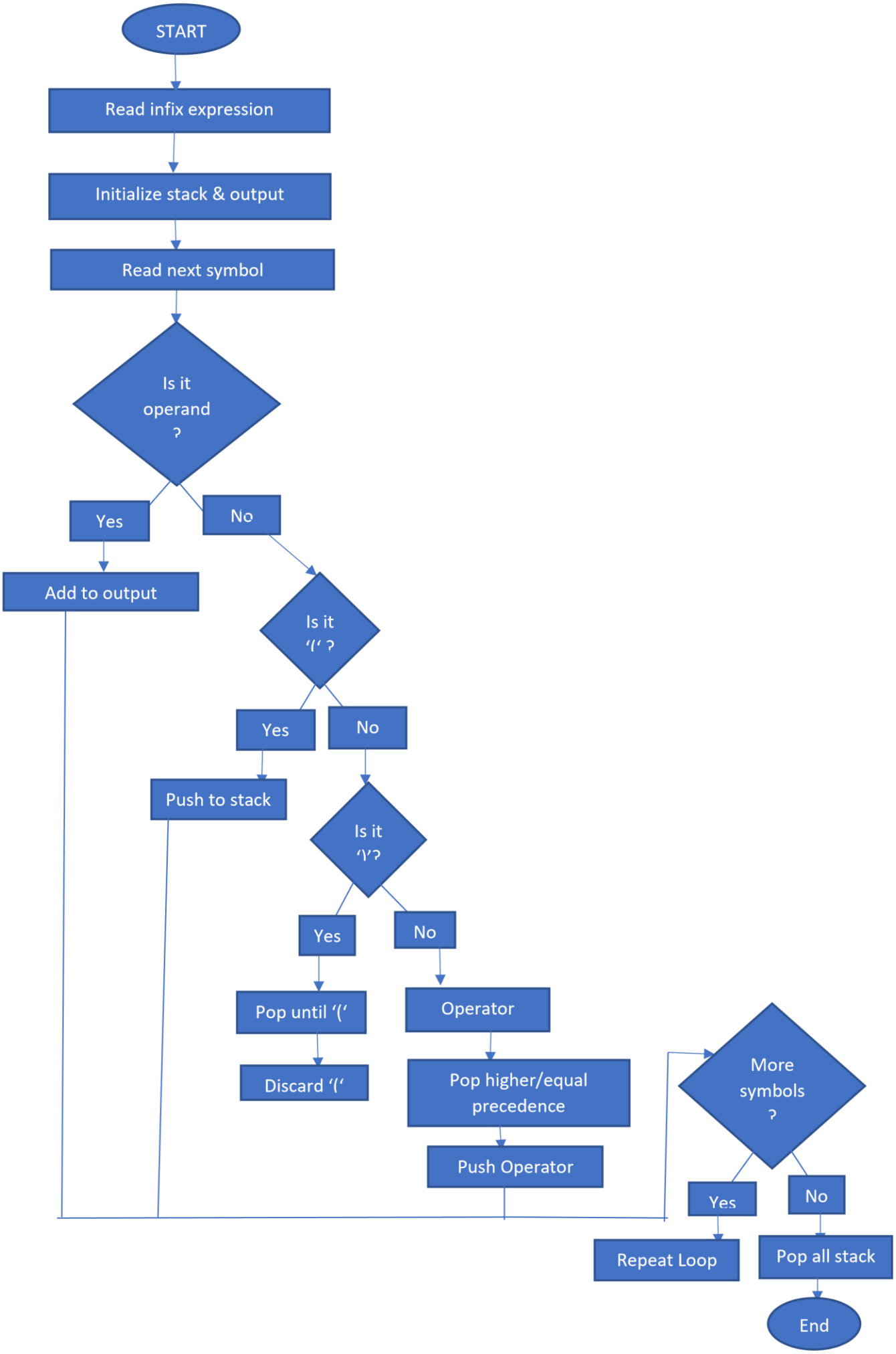
DUE DATE: 16TH APRIL 2026

FLOWCHARTS

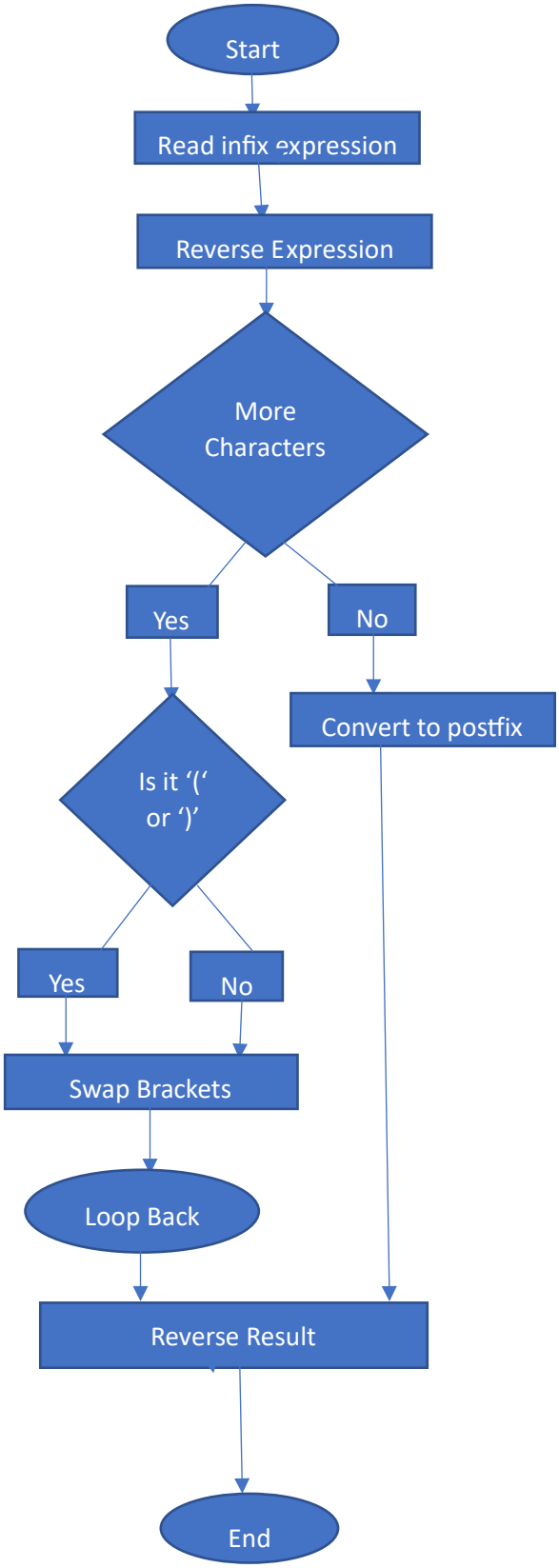
INFIX TO POSTFIX

FLOWCHART

INFIX TO POSTFIX


START
Read infix expression
Initialize stack & output
Read next symbol
Is it operand ?
Yes
No
Add to output
Is it '(' ?
Yes
No
Push to stack
Is it ')'?
Yes
No
Pop until '('
Discard '('
Operator
Pop higher/equal precedence
Push Operator
More symbols ?
Yes
No
Repeat Loop
Pop all stack
End

INFIX TO PREFIX



PSEUDO CODES

1.INFIX TO POSTFIX function

infixToPostfix(expression):

create empty stack result = "" for each character ch in expression: if ch is operand: add ch to result else if ch == '(': push ch to stack else if ch == ')': while top of stack != '(': pop and add to result pop '(' from stack else: while stack not empty AND precedence(top) >= precedence(ch):

pop and add to result push ch to stack while stack not empty: pop and add to result return resultfunction infixToPrefix(expression):

reverse expression

for each character:

if '(' replace with ')'

if ')' replace with '('

postfix = infixToPostfix(reversed expression)

reverse postfix

return prefix

2.<u>INFIX TO PREFIX</u>

function infixToPrefix(expression):

reverse expression

for each character:

if '(' replace with ')'

if ')' replace with '('

postfix = infixToPostfix(reversed expression)

reverse postfix return prefix

JAVA PROGRAM

```
import java.util.Stack; public class
InfixConversion { // Check
precedence static int
precedence(char ch) { if (ch == '+'
|| ch == '-') return 1; if (ch == '*'
|| ch == '/') return 2; return -1;
  }

  // Check operand static boolean
isOperand(char ch) { return
Character.isLetterOrDigit(ch);
  }

  // Infix to Postfix
  static String infixToPostfix(String exp) {
    StringBuilder result = new StringBuilder();
    Stack<Character> stack = new Stack<>();

    for (char ch : exp.toCharArray()) {

      if (isOperand(ch)) {
result.append(ch);
      }

      else if (ch == '(') {
stack.push(ch);
      }
```

```
        else if (ch == ')') {
while (stack.peek() != '(') {
result.append(stack.pop());
            }
            stack.pop();
        }
else {
while
(!stack.isEm
pty() &&
                precedence(stack.peek()) >= precedence(ch)) {
result.append(stack.pop());
            }
            stack.push(ch);
        }
    }

    while (!stack.isEmpty()) {
result.append(stack.pop());
    }

    return result.toString();
  }

  // Reverse string     static String
reverse(String str) {
    return new StringBuilder(str).reverse().toString();
```

```
    }

    // Infix to Prefix
    static String infixToPrefix(String exp) {
// Step 1: Reverse
        String reversed = reverse(exp);

        // Step 2: Swap brackets
        StringBuilder modified = new StringBuilder();
for (char ch : reversed.toCharArray()) {
            if (ch == '(')
modified.append(')');            else if
(ch == ')')
modified.append('(');            else
                modified.append(ch);
        }

        // Step 3: Convert to postfix
        String postfix = infixToPostfix(modified.toString());

        // Step 4: Reverse postfix
return reverse(postfix);
    }

    public static void main(String[] args) {
        String infix = "A+B*C-D/F";

        System.out.println("Infix: " + infix);
```

```
        System.out.println("Postfix: " + infixToPostfix(infix));
        System.out.println("Prefix: " + infixToPrefix(infix));
    }
}
```

```
C:\WINDOWS\system32\cmd.

Microsoft Windows [Version 10.0.26100.8117]
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C:\Users\HC ELETRONICS>CD DESKTOP

C:\Users\HC ELETRONICS\Desktop>javac InfixConversion.java

C:\Users\HC ELETRONICS\Desktop>java InfixConversion
Infix: A+B*C-D/F
Postfix: ABC*+DF/-
Prefix: +A-*BC/DF

C:\Users\HC ELETRONICS\Desktop>
```